

ate
106/26

(Sub-D.S.A)

week-1 module-programming practice
(3rd year)

Star patterns

① Square

code:-

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter no. of rows";
    cin >> n;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            cout << "*";
        }
        cout << endl;
    }
}
```

logic:-

- Here we use only $(i=j)$ concept.

Row = column

- Here we give the value of (n) by ourselves.

output

$n = 5$

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

② Hollow square

```
#include <iostream>
using namespace std;
int main() {
    int n = 5;
```

```
for(int i=0; i<n; i++)
```

```
{
```

```
for(int j=0; j<n; j++)
```

```
{
```

```
if(i==0 || j==0 || i==n-1 || j==n-1)
```

```
{
```

```
cout << "X";
```

```
} else {
```

```
cout << " ";
```

```
}
```

```
cout << endl;
```

```
}
```

```
}
```

```
}
```

Logic,

- use just use the same loop as above

but apply the condition that first & last line are same but middle is hollow with $n-1$

output :-

```
X X X X X
```

```
X       X
```

```
X       X
```

```
X       X
```

```
X X X X X
```

③ Triangle

code

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
int n;
```



```

for (int i=1; i<=n; i++) {
    for (int j=1; j<=i; j++) {
        cout << "★";
    }
    cout << endl;
}
}

```

Output

```

★
★★
★★★
★★★★
★★★★★

```

④ Inverted Triangle

```

★★★★★
★★★★
★★★
★★
★

```

code

```

int main() {
    int n;
    for (int i=1; i<=n; i++) {
        for (int j=1; j<=n+1-i; j++) {
            cout << "★";
        }
        cout << endl;
    }
}

```

logic

- Same as above but only one change is $(j <= n+1-i)$.

logic

- using for loop $(j <= i)$ is the main logic here.

⑤ pyramid.

```
  *
 * * *
* * * * *
* * * * * * *
* * * * * * * * *
```

code

```
int main() {
    int n;
    cout << "enter the
    no. of rows";
    cin >> n;
    for (int i = 1; i <= n; i++)
    {
        for (int j = 1; j <= n - i; j++)
            cout << " "; // space
```

```
    }
    for (int k = 1; k <= 2 * i - 1; k++) {
        cout << "*";
    }
    cout << endl;
}
```

logic

- Here what we take the value of $n=0$ in starting.
- 'first loop' for the space and in the second we use this for the required output to be print.

⑥ Diamond

A large pyramid of asterisks is drawn on lined paper. The pyramid consists of 10 rows of asterisks, with the number of asterisks in each row decreasing from 10 at the top to 1 at the bottom. The asterisks are arranged in a symmetrical, triangular pattern. Below the pyramid, there is a single asterisk centered on the line.

logic.

- $(2n-1)$ line print $\rightarrow$ outer loop.
- $usp = n-1 \rightarrow usp-- / usp++$
- $ust = 1, ust += 2 / ust -= 2$
 $\swarrow$ firstly $\searrow$ then
After

code :-

```

int main() {
    int n;
    cout << "NO. of line: ";
    cin >> n;
    int usp = n - 1;
    int usft = 1;
    for (int i = 1; i <= 2 * n - 1; i++) {
        for (int j = 1; j <= usp; j++) {
            cout << " ";
        }
        if (i <= n - 1) usp--;
        else usft++;
        for (int k = 1; k <= usft; k++) {
            cout << "★";
        }
        if (i <= n - 1) usft += 2;
        else usft -= 2;
        cout << endl;
    }
}

```

⑦ Half Diamond

```
*
**
***
****
*****
****
***
**
*
```

logic

* print upper half of the above pattern as like triang ③. pattern.

* print lower half of the pattern as like inverted triangle (starting i for n-1).

code :-

```
int main() {
    int n=5;
    for(int i=1; i<=n; i++) {
        for(int j=1; j<=i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    for(int i=n-1; i>=1; i--) {
        for(int j=1; j<=i; j++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

⑧ Half Diamond Inverted.

logic

- print stars from n down to 1.
- then print stars from 2 up to n .
- use the loop.

code

```
#include <iostream>
using namespace std;
int main () {
    int n = 5;
    for (int i = n; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            cout << "x";
        }
        cout << endl;
    }
    for (int i = 2; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "x";
        }
        cout << endl;
    }
    return 0;
}
```

Output

x x x x x

x x x x

x x x

x x

x

x x

x x x

x x x x

x x x x x

first part of loop

second part of loop

2. Number pattern

① square

```
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
```

logic

- same as the star pattern.
- use for loop in which $(i=j)$

code

```
#include <iostream>
using namespace std;
int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            cout << "1";
        }
        cout << endl;
    }
}
```

② Incrementing

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

logic

- same as above.
- outer & the inner loop $(1 \text{ to } 5)$

code

```
int main() {  
    int n = 5;  
    for (int i = 1; i <= n; i++) {  
        for (int j = 1; j <= n; j++) {  
            cout << j << " ";  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

③ Right Triangle

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

Logic

• outer loop $\rightarrow (1 \text{ to } n)$

• inner loop $\rightarrow (1 \text{ to the current row number})$

code :-

```
int main() {  
    int n = 5;  
    for (int i = 1; i <= n; i++) {  
        for (int j = 1; j <= i; j++) {  
            cout << j << " ";  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

④ Inverted Diamond

3 3 3 3 3

4 4 4 4

5 5 5

6 6

7

logic

- start number from 3.
- outer loop controls rows (1 to 5)

- in each row :-

→ print the current no. (num)
 $n - i + 1$

same as inverted
 triangle but
 only
 number change

→ increase karo num by 1
 for next row.

code

```
int main() {
```

```
    int n = 5;
```

```
    int num = 3;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        for (int j = 1; j <= n - i + 1; j++)
```

```
        {
```

```
            cout << num << " ";
```

```
        }
```

```
        num++;
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```

⑤ Sandwich pattern

3 3 3 3 3

4 4 4

5

4 4 4

3 3 3 3 3

logic

• upper part

↳ Basically we have to print the 3 five times, 4 three times & 5 one times.

• same as in the lower part we have to do is that.

↳ print 4 three times

↳ print 3 five times.

code :-

```
int main() {
    for (int i=0; i<3; i++) {
        for (int s=0; s<i; s++)
            cout << " ";
        for (int j=0; j<5-2*i; j++)
            cout << i+3 << " ";
        cout << endl;
    }
    for (int i=1; i>=0; i--) {
        for (int s=0; s<i; s++)
            cout << " ";
        for (int j=0; j<5-2*i; j++)
            cout << i+3 << " ";
        cout << endl;
    }
    return 0;
}
```

DRY RUN

$i = 0 \rightarrow 33333$

$i = 1 \rightarrow 444$

$i = 2 \rightarrow 5$

$i = 1 \rightarrow 444$

$i = 0 \rightarrow 3333$

(6) Incrementing Diamond

3

4 4

5 5 5

6 6 6 6

7 7 7 7 7

6 6 6 6

5 5 5

4 4

3

logic

• Same as a triangle concept.

• Assume two part of the triangle and then solve them.

code

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n = 5;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        for (int j = 1; j <= i; j++) {
```

```
            cout << i + 2 << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    for (int i = n - 1; i >= 1; i--) {
```

```
        for (int j = 1; j <= i; j++) {
```



```
cout << endl;
```

```
3
```

```
return 0;
```

```
2
```

Square.cpp > main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n;
5      cout<<"Enter the no of rows :";
6      cin>>n;
7      for(int i=1; i<=n; i++){
8          for(int j=1; j<=n; j++){
9              cout<<"*";
10
11          }
12      cout<<endl;
13  }
14 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> cd .
) { g++ Square.cpp -o Square } ; if ($?) { .\Square }
```

Enter the no of rows :5

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module>
```



```

HollowSquar.cpp > main()
1  // HOLLOW SQUARE //
2  #include<iostream>
3  using namespace std;
4  int main(){
5      int n=5;
6      for(int i=0; i<n; i++){
7          for(int j=0; j<n; j++){
8              if(i==0||j==0||i==n-1||j==n-1){
9                  cout<<"*";
10             }
11             else{
12                 cout<<" ";
13             }
14         }
15         cout<<endl;
16     }
17 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

> cd "c:\Users\Utsaw
) { g++ HollowSquar.cpp -o HollowSquar } ; if ($?) { .\HollowSquar }

```

```
*****
```

```
*   *
```

```
*   *
```

```
*   *
```

```
*****
```

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module>
```

Triangle.cpp > main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n;
5      cout<<"Enter the no of rows :";
6      cin>>n;
7      for(int i=1; i<=n; i++){
8          for(int j=1; j<=i; j++){
9              cout<<"*";
10         }
11         cout<<endl;
12     }
13 }
14 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

```
PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module> cd "c:\U
) { g++ Triangle.cpp -o Triangle } ; if ($?) { .\Triangle }
```

Enter the no of rows :5

*

**

```
PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module> 
```


G++ InvertedTriangle.cpp > main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n;
5      cout<<"Enter the no of rows :";
6      cin>>n;
7      for(int i=1; i<=n; i++){
8          for(int j=1; j<=n+1-i; j++){ // here concept for the reversal
9              cout<<"*";
10         }
11         cout<<endl;
12     }
13 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> cd "c:\Users\Utsaw kumar\OneDrive\
Desktop\week 1 Module" & { g++ InvertedTriangle.cpp -o InvertedTriangle } ; if ($?) { .\InvertedTriangle }
```

Enter the no of rows :5

**

*

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> 
```

G++ InvertedTriangle.cpp > main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n;
5      cout<<"Enter the no of rows :";
6      cin>>n;
7      for(int i=1; i<=n; i++){
8          for(int j=1; j<=n+1-i; j++){ // here concept for the reversal
9              cout<<"*";
10         }
11         cout<<endl;
12     }
13 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> cd "c:\Users\Utsaw kumar\OneDrive\
Desktop\week 1 Module" & { g++ InvertedTriangle.cpp -o InvertedTriangle } ; if ($?) { .\InvertedTriangle }
```

Enter the no of rows :5

**

*

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> 
```


 Diamond.cpp >  main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n;
5      cout<<"No of line :";
6      cin>>n;
7      int nsp=n-1;
8      int nst=1;
9      for(int i=1;i<=2*n-1;i++){
10         for(int j=1;j<=nsp;j++){
11             cout<<" ";
12         }
13         if(i<=n-1) nsp--;
14         else nsp++;
15         for(int k=1;k<=nst;k++){
16             cout<<"*";
17         }
18         if(i<=n-1)nst+=2;
19         else nst-=2;
20         cout<<endl;
21     }
22 }
```

```
> c
) { g++ Diamond.cpp -o Diamond } ; if ($?) { .\Diamond }
```

```
No of line :5
```

```
    *
```

```
    ***
```

```
    *****
```

```
    *********
```

```
    ***********
```

```
    *********
```

```
    *****
```

```
    ***
```

```
    *
```

```
PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module>
```



```

pyramid.cpp > main()
2  #include<iostream>
3  using namespace std;
4  int main(){
5      int n;
6      cout<<"Enter the no of rows :";
7      cin>>n;
8      for(int i=1; i<=n; i++){
9          for(int j=1; j<=n-i; j++){
10             cout<<" ";
11         }
12         for(int k =1;k<=2*i-1;k++){
13             cout<<"*";
14         }
15         cout<<endl;
16     }
17     return 0;
18 }
19

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

> cd "c:\Users\Utsav k
) { g++ pyramid.cpp -o pyramid } ; if ($?) { .\pyramid }
Enter the no of rows :5

```

```

*
***
*****
*****
*****

```

PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module>

Ln 10, Col 21 Spaces

HalfDiamond.cpp > main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n=5;
5      for(int i=1; i<=n;i++){
6          for(int j=1; j<=i;j++){
7              cout<<"*";
8          }
9          cout<<endl;
10     }
11     for(int i=n-1; i>=1;i--){
12         for(int j=1;j<=i; j++){
13             cout<<"*";
14         }
15         cout<<endl;
16     }
17     return 0;
18 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

```
> cd "c:\Users\
) { g++ HalfDiamond.cpp -o HalfDiamond } ; if ($?) { .\HalfDiamond }
*
**
***
****
*****
*****
****
***
**
*
```

PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module>

InvertedTriangle.cpp

pyramid.cpp

Diamond.cpp

HalfDiamond.cpp

halfdiamondinverted.cpp > main()

```
1  #include<iostream>
2  using namespace std ;
3  int main(){
4      int n=5 ;
5      for(int i=n;i>=1;i--){
6          for(int j=1;j<=i;j++){
7              cout<<"*";
8          }
9          cout<<endl;
10     }
11     for(int i=2;i<=n;i++){
12         for(int j=1;j<=i;j++){
13             cout<<"*";
14         }
15         cout<<endl;
16     }
17     return 0;
18 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> cd "c:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module"

) { g++ halfdiamondinverted.cpp -o halfdiamondinverted } ; if (\$?) { .\halfdiamondinverted.exe }

**

*

**

PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module>

number >  square.cpp >  main()

```
1  #include<iostream>
2  using namespace std;
3  ∨ int main(){
4      int n =5;
5
6  ∨      for(int i=1; i<=n; i++){
7  ∨          for(int j=1; j<=n; j++){
8              cout<<"1";
9
10             }
11             cout<<endl;
12         }
13     }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

```
PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module> cd "c:\Us
if ($?) { g++ square.cpp -o square } ; if ($?) { .\square }
```

```
11111
```

```
11111
```

```
11111
```

```
11111
```

```
11111
```

```
PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module\number>
```

number >  increasementing.cpp >  main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n =5;
5
6      for(int i=1; i<=n; i++){
7          for(int j=1; j<=n; j++){
8              cout<<j<<" ";
9          }
10         cout<<endl;
11     }
12 }
13 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> cd "c:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module"
if ($?) { g++ increasementing.cpp -o increasementing } ; if ($?) { .\increasementing }
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module\number>
```


number >  righttriangle.cpp >  main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n =5;
5
6      for(int i=1; i<=n; i++){
7          for(int j=1; j<=i; j++){
8              cout<<j<<" ";
9
10             }
11         cout<<endl;
12     }
13 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module> cd "c:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module"

if (\$?) { g++ righttriangle.cpp -o righttriangle } ; if (\$?) { .\righttriangle.exe }

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module\number>

number >  inverteddiamond.cpp >  main()

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      int n=5;
5      int num=3;
6      for(int i=1; i<=n; i++){
7          for(int j=1; j<=n-i+1; j++){
8              cout<<num<<" ";
9          }
10         num++;
11         cout<<endl;
12     }
13     return 0;
14 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> cd "c:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module\number\" ; if ($?) { g++ inverteddiamond.cpp -o inverteddiamond }
```

```
3 3 3 3 3
```

```
4 4 4 4
```

```
5 5 5
```

```
6 6
```

```
7
```

```
PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module\number>
```

[square.cpp](#)[increasementing.cpp](#)[righttriangle.cpp](#)number > [sandwichpatrren.cpp](#) >  main()

std;

Click to add a breakpoint

```
4      for(int i=0; i<3; i++){
5          for(int s=0;s<i;s++)
6              cout<<" ";
7          for(int j=0;j<5-2*i;j++)
8              cout<<i+3<<" ";
9              cout<<endl;
10         }
11         for(int i=1;i>=0;i--){
12             for(int s=0;s<i;s++)
13                 cout<<" ";
14             for(int j=0;j<5-2*i;j++)
15                 cout<<i+3<<" ";
16             cout<<endl;
17         }
18     return 0;
19 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module> cd "c:\U

if (\$?) { g++ sandwichpatrren.cpp -o sandwichpatrren } ; if (\$?)

3 3 3 3 3

4 4 4

5

4 4 4

3 3 3 3 3

PS C:\Users\Utsaw kumar\OneDrive\Desktop\week 1 Module\number>

number >  incrementingdiamond.cpp >  main()

Click to add a breakpoint  am>

```
1 // #include <iostream>
2 // using namespace std;
3 int main(){
4     int n=5;
5     for(int i=1;i<=n;i++){
6         for(int j=1; j<=i; j++){
7             cout<<i+2<<" ";
8         }
9         cout<<endl;
10    }
11    for(int i=n-1;i>=1;i--){
12        for(int j=1;j<=i;j++){
13            cout<<i+2<<" ";
14        }
15        cout<<endl;
16    }
17    return 0;
18 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module> cd "c:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module"

if (\$?) { g++ incrementingdiamond.cpp -o incrementingdiamond } ; if (\$?) {

3

4 4

5 5 5

6 6 6 6

7 7 7 7 7

6 6 6 6

5 5 5

4 4

3

PS C:\Users\Utsav kumar\OneDrive\Desktop\week 1 Module\number>